Quiz 3
Wireless Networks

Q1. Suppose you are using multipath TCP. Your NIC acts as segment-splitting middlebox and in addition, changes the initial sequence number. How would multipath TCP work in this scenario? Explain in not more than 4-5 sentences. [1.5+ 1.5]

Q2. Suppose you are performing web browsing over wireless. The HTTP version is HTTP/1.1. The browser opens 6-8 multiple parallel TCP connections to the same web server. We observed that HTTP is chatty and hence increases uplink traffic and hence collision on the channel. Suppose, without changing the HTTP protocol, we used MPTCP here i.e., instead of 6-8 connections the transport protocol creates only a single TCP connection and adds these 6-8 connections as subflows of the same connection. Would this reduce uplink traffic and result in lesser contention/collision in a WiFi network? Answer in not more than two sentences. [2]

Q5. Marks the one(s) FALSE [2]

a. Split-TCP does not maintain end-to-end semantics
b. Snoop TCP does not maintain end-to-end semantics
c. Snoop needs to maintain state at the BS
d. ECN is sent when congestion occurs

Q3. Suppose you are using WiFi CSI to perform three different activity recognition: stationary, walking, and running. Suppose, you collected a dataset in C21 of old acad i.e., our own classroom during class hours. Now, you trained a model for this. Next, you want to use the same model in the following environments. Tell whether the accuracy will improve/remain the same/ will degrade and also tell the reason. [2 + 2]

a. In the canteen during lunchtime
b. In C21 when nobody is present

Q4. Suppose, a multipath TCP sender uses EWTCP as the congestion control algorithm and uses a value of alpha=1/k and k=4. [2 +2]

a. Will it be fair to regular TCP? Explain your answer
b. Will it be more efficient than regular TCP? Explain your answer.